Exploring improvements of quantum-enhanced Markov chain Monte Carlo

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Sampling from Boltzmann distributions is crucial in scientific computing with applications ranging from statistical physics to machine learning. Recently, *Layden et al.* [Nature 619, 282–287 (2023)] proposed a quantum-enhanced Markov chain Monte Carlo (MCMC) algorithm that uses quantum dynamics to propose transitions between classical states. While this algorithm showed an empirical polynomial speedup, the physical origin of the speedup remains unclear. We analyze the parameter regimes of quantum-enhanced MCMC for optimal performance and conjecture that the observed speedup is connected to the phase transition of the quantum spin glass. Additionally, we explore alternative quantum proposal strategies using classically optimized time-dependent Hamiltonian evolution. Looking forward, we expect that our empirical findings will aid future theoretical studies in understanding the precise mechanisms underlying the quantum speedup.

Talk outline

1. Sampling and MCMC basics
2. Quantum-enhanced MCMC
3. Optimal parameter regimes
4. Alternative approaches
5. Big picture view

Questions are welcome!

Sampling

$\pi : S \to [0, 1]$

Boltzmann: $\pi(s) = \frac{e^{-\beta H(s)}}{\mathcal{Z}}$

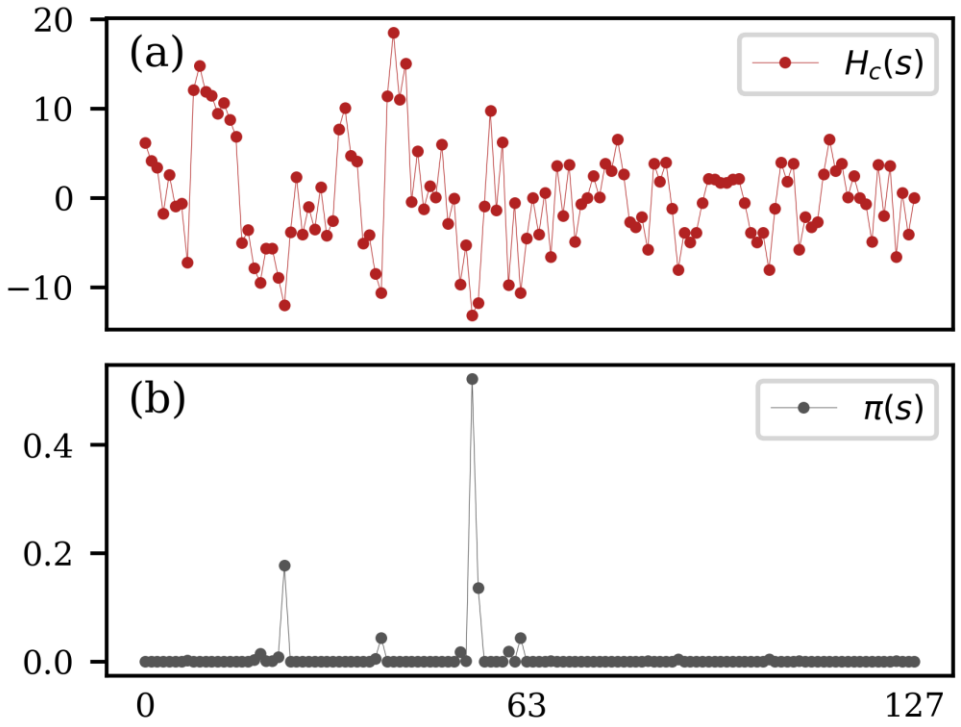
where $\mathcal{Z} = \sum_s e^{-\beta H(s)}$

State: $s \in \{\pm 1\}^n$

Fields and couplings: $J_{ij}, h_i \sim \mathcal{N}(0, 1)$

Spin-glass Hamiltonian:

$$H_c(s) = -\frac{1}{2} \sum_{i,j=1}^{n} J_{ij} s_i s_j - \sum_{i=1}^{n} h_i s_i$$



MCMC Basics

(1) Propose new state with $Q(s|s_0)$

(2) Accept the proposed state with $A(s|s_0)$

Metropolis-Hastings:

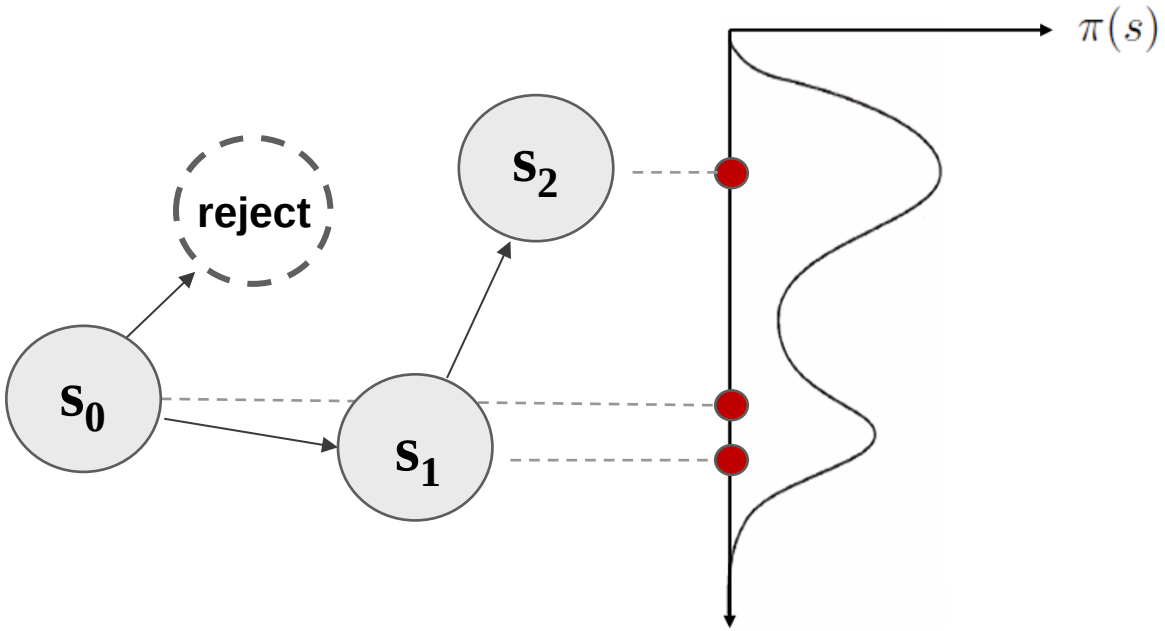
$$A(s|s_0) = \min\left(1, \frac{\pi(s)Q(s_0|s)}{\pi(s_0)Q(s|s_0)}\right)$$

Transition matrix: $P(s_0 \to s) = Q(s|s_0)A(s|s_0)$

Spectral gap: $\delta = 1 - \max_{i \neq 1} |\lambda_i|$

where $1 = \lambda_1 \geq \lambda_2 \geq \lambda_3 \geq \ldots \geq \lambda_d$ are the eigenvalues of P

Mixing time: $(\delta^{-1} - 1)\ln\left(\frac{1}{2\epsilon}\right) \leq t_{\text{mix}} \leq \delta^{-1}\ln\left(\frac{1}{\epsilon\ \min_s \pi(s)}\right)$



Quantum proposal

Local: Flip one spin in the current state randomly

Random: Propose new state uniformly at random

Quantum: $Q(s|s_0) = |\langle s|U|s_0\rangle|^2$

Symmetry: $U = U^T \implies A(s|s_0) = \min\left(1, e^{-\beta\Delta H_c}\right)$

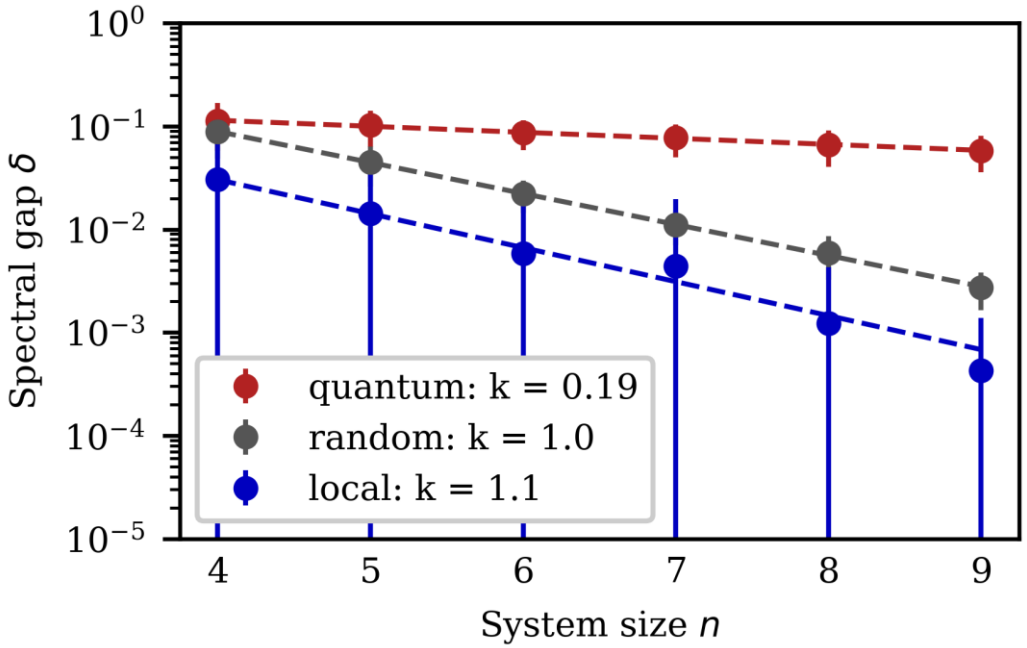
Choice of U: $U = e^{-iHt}$

Hamiltonian = Spin glass + mixing term

$$H = (1-\gamma)\alpha H_c + \gamma H_{mix}$$

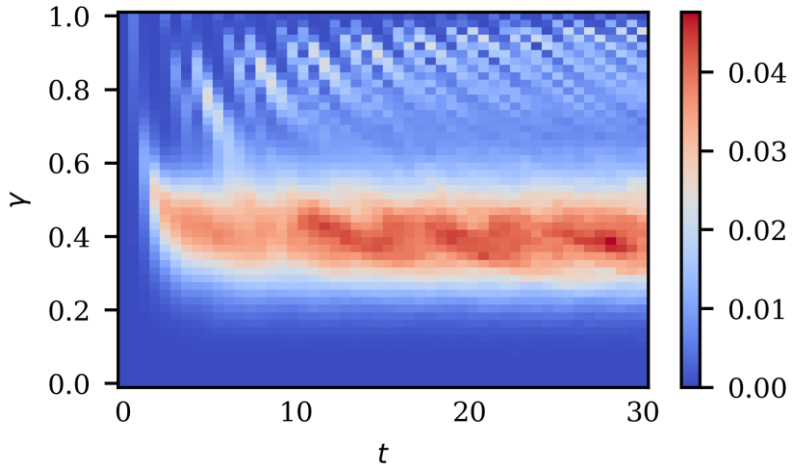
where $H_{mix} = \sum_{i=1}^{n} X_i$

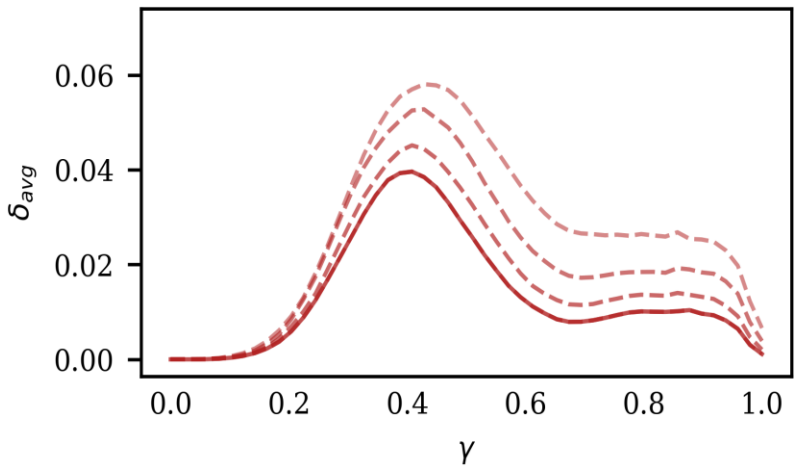
Scaling with system size: $\delta \propto 2^{-kn}$



Parameter grid search

$$H = (1-\gamma)\alpha H_c + \gamma H_{mix}$$

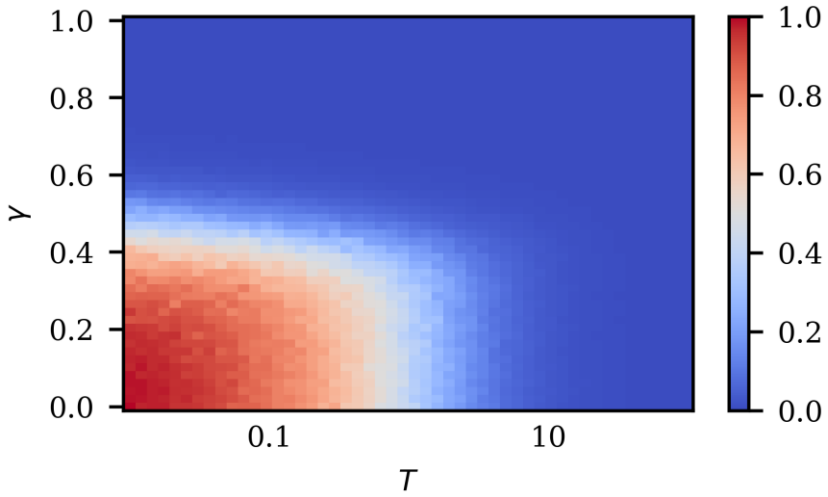




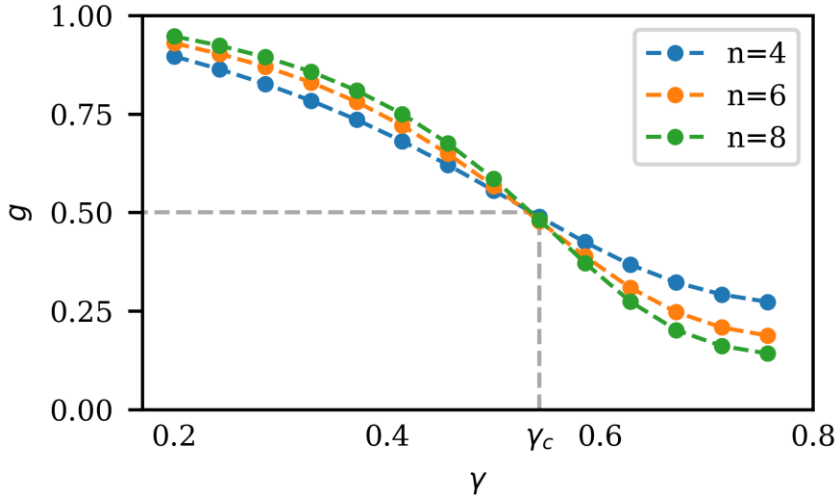
Large γ: $U = e^{-iH_{mix}t} = \prod_{i=1}^{n} e^{-iX_i t}$

Small γ: $P(|j\rangle \to |k\rangle) \propto |\langle k|H_{mix}|j\rangle|^2\gamma^2 + \mathcal{O}(\gamma^3)$

Locating critical γ value



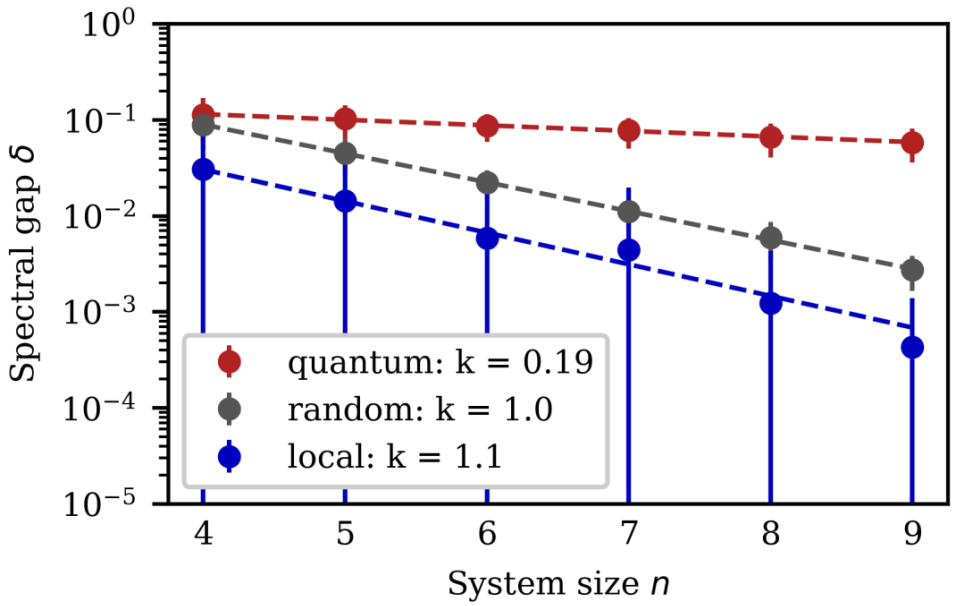
$$\gamma_c = 0.55 \pm 0.04$$

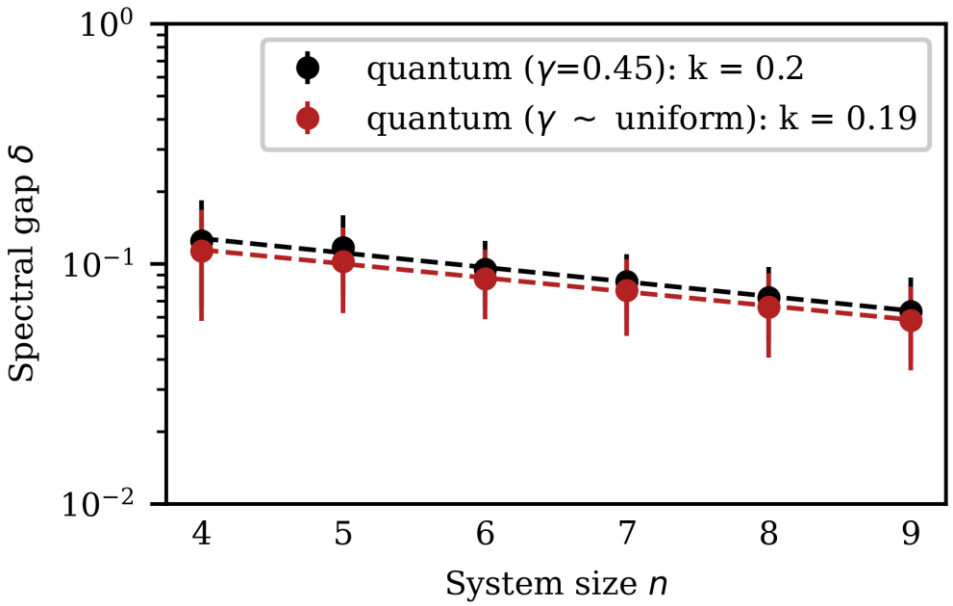


Order parameter: $q = \frac{1}{n}\sum_{i=1}^{n}\overline{\langle\sigma_i\rangle^2}$ (Edward-Anderson)

Finite size analysis: $g = \frac{1}{2}\left[3 - \overline{\left(\frac{q^{(4)}}{(q^{(2)})^2}\right)}\right]$ (Binder cumulant)

Fixed near-critical γ

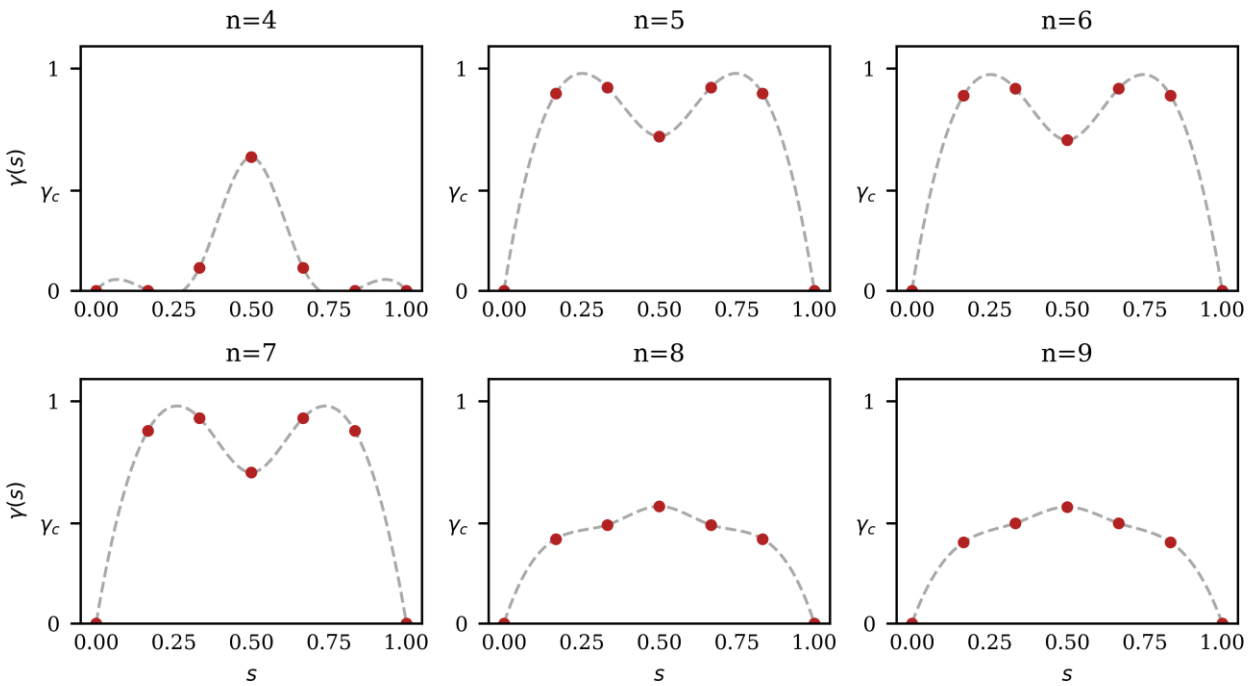




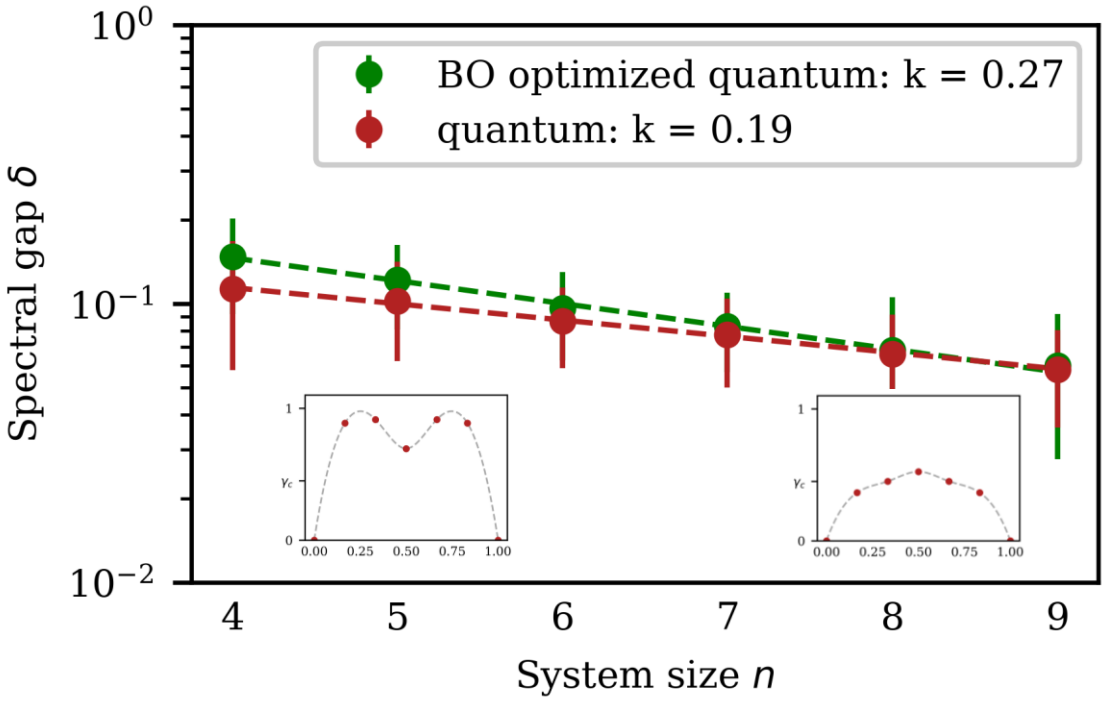
Classically optimized schedules

Time-dependent Hamiltonian: $H(t) = (1 - \gamma(t))\alpha H_c + \gamma(t) H_{mix}$

What is the optimal $\gamma(t)$ shape? Let the classical optimizer find the optimal shape.



Performance of optimized schedules

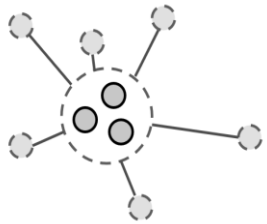


Big picture

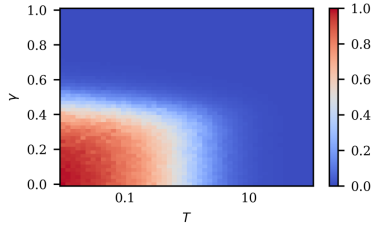
MCMC convergence

δ

Bottleneck analysis

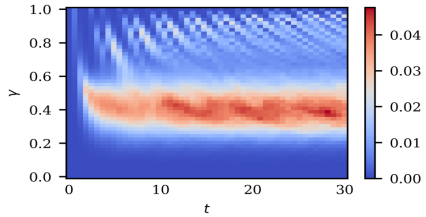


Phase transitions



Physics

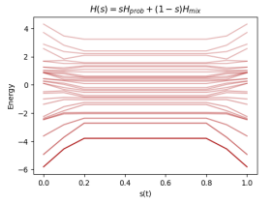
Empirical analysis



Population flow

$$P_D = 1 - e^{-2\pi\Gamma}$$

$$\Gamma = \frac{a^2/\hbar}{\left|\frac{\partial}{\partial t}(E_2 - E_1)\right|}$$

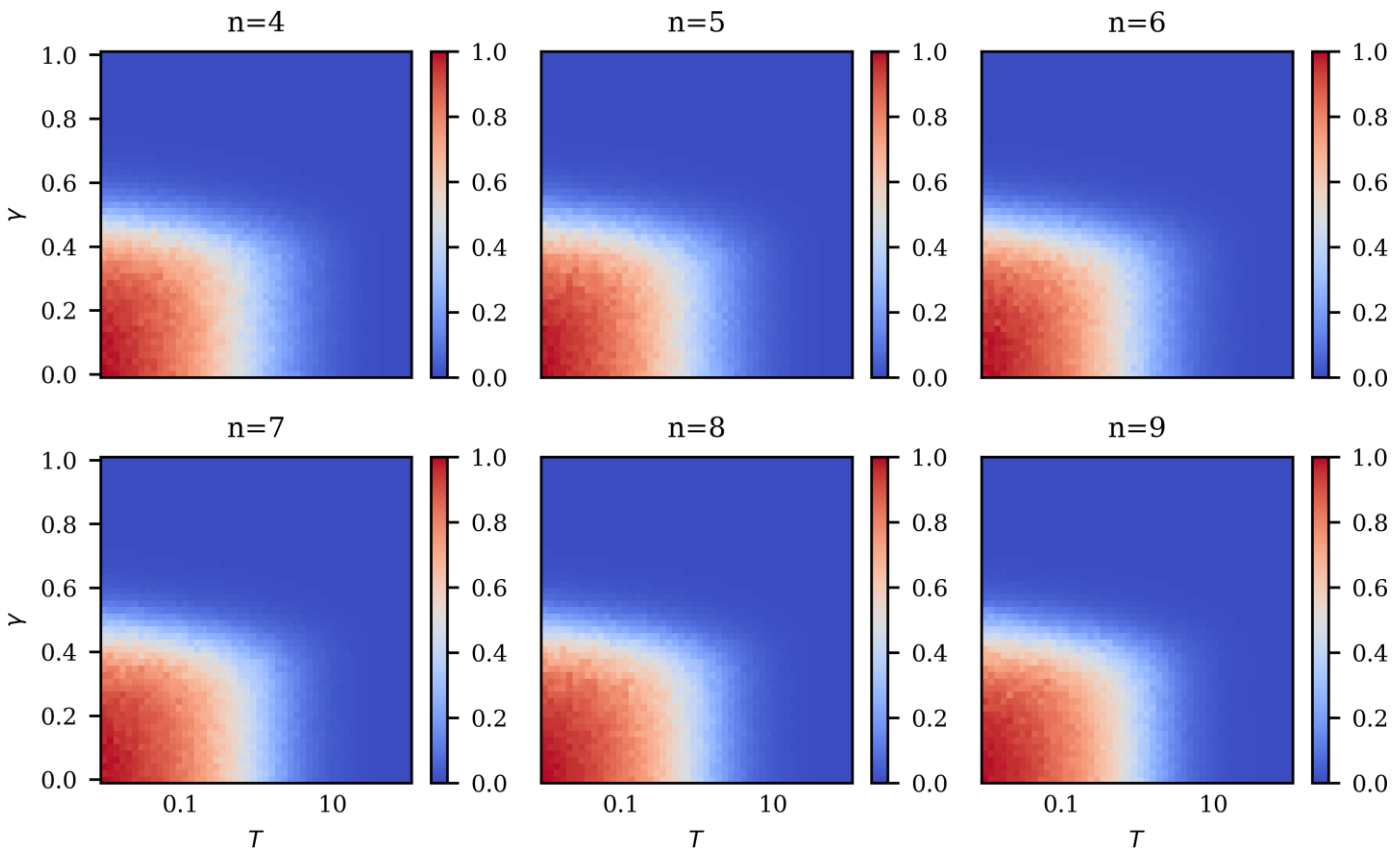


Open question

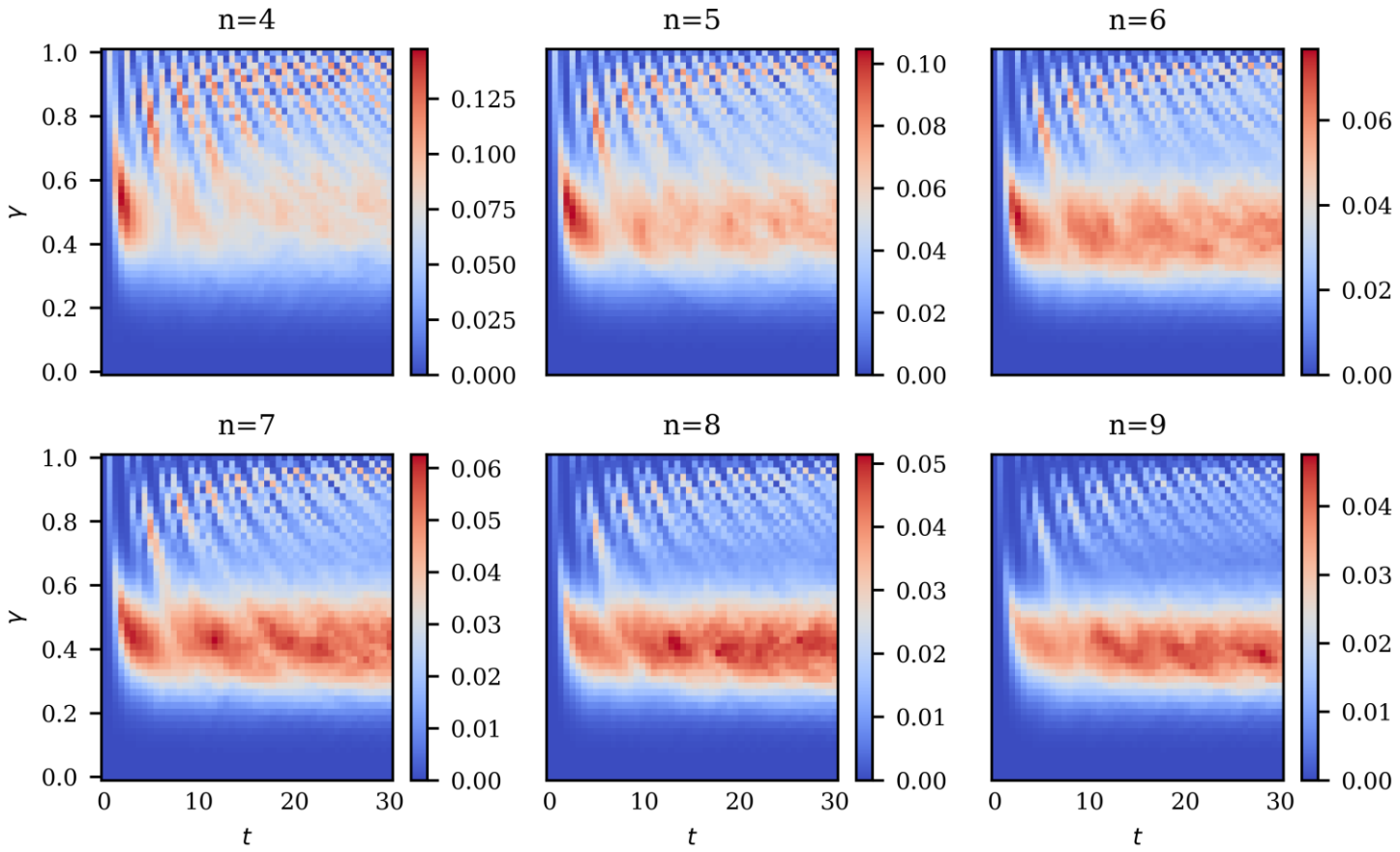
1. Does the polynomial speedup persist for larger systems?
2. How hard is it to simulate the quantum proposal classically?
3. Can we connect convergence rate with physical properties of the system?
4. What kind of problems is quantum-enhanced MCMC good for?

Backup slides

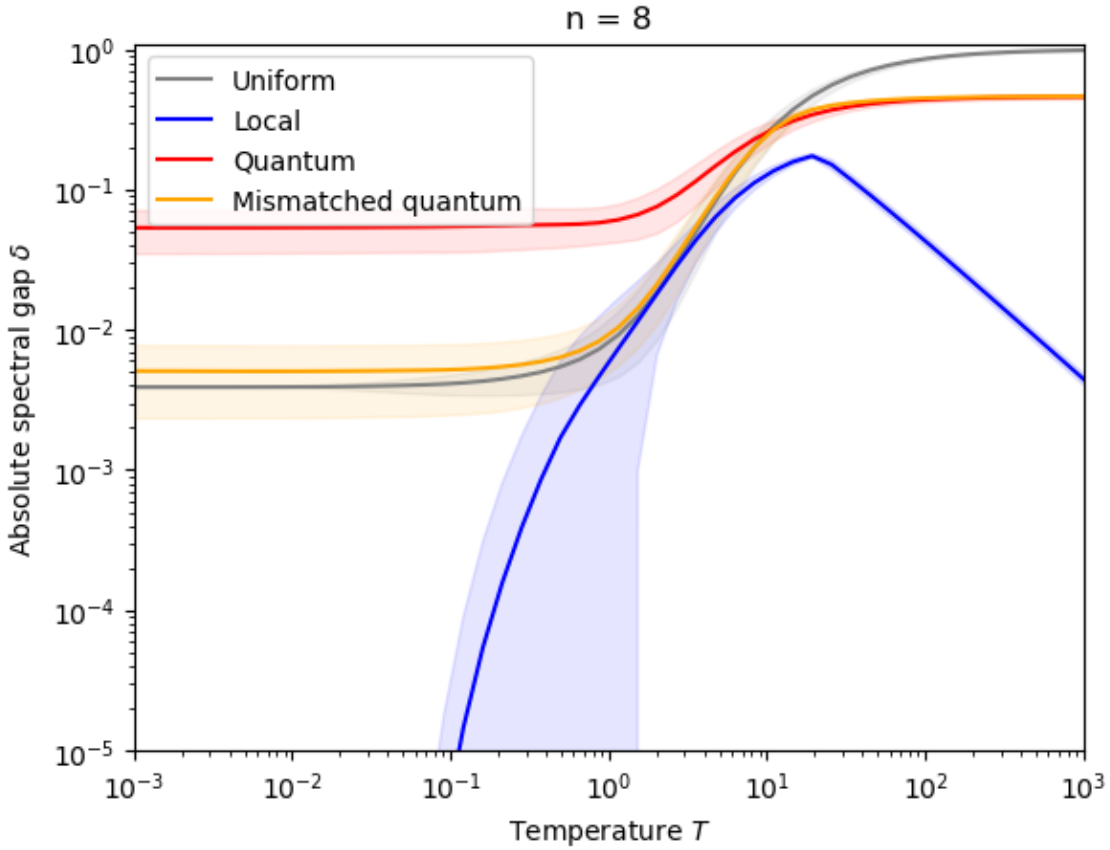
Phase diagrams for different **n**



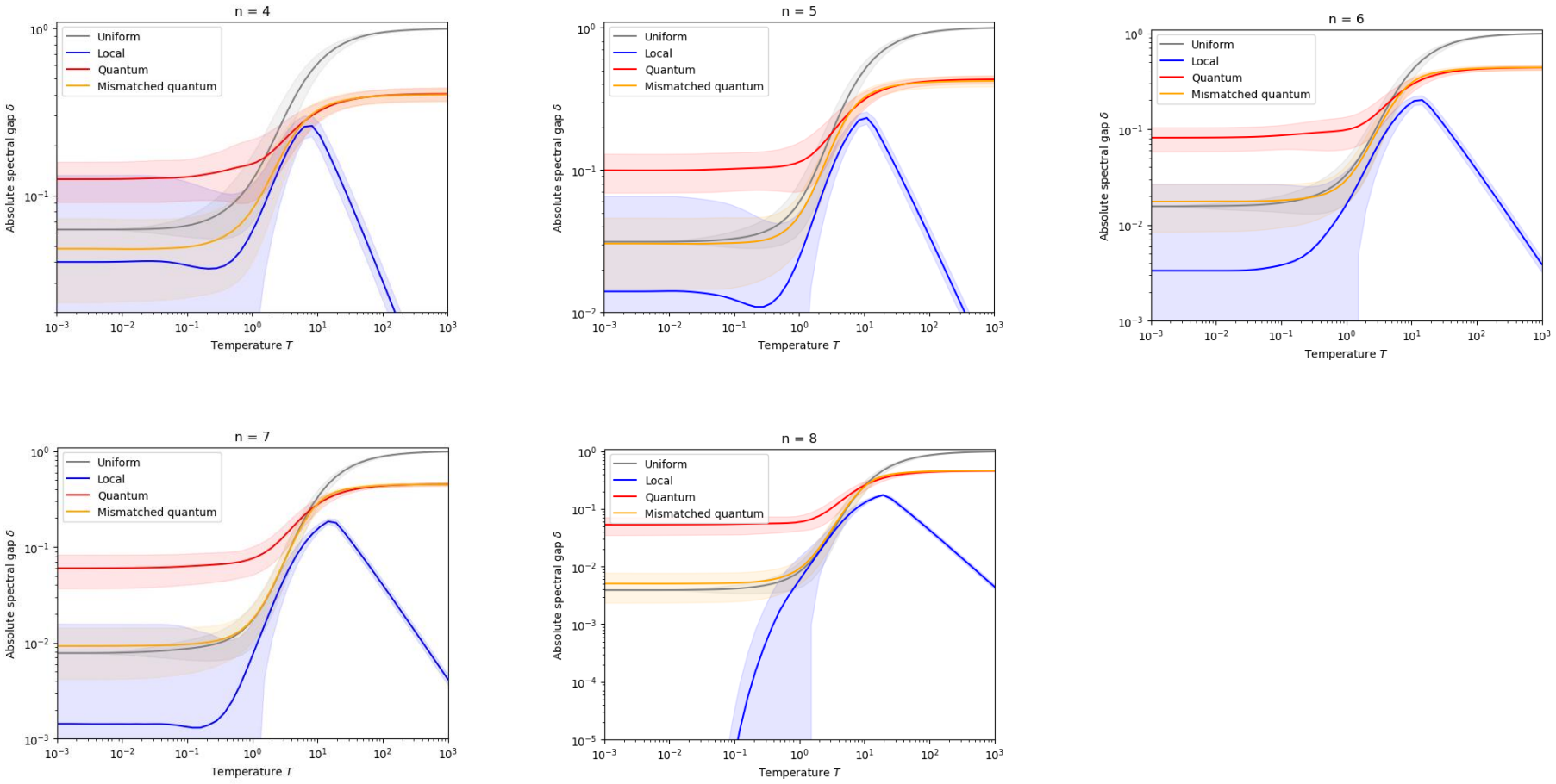
Parameter landscape for different n



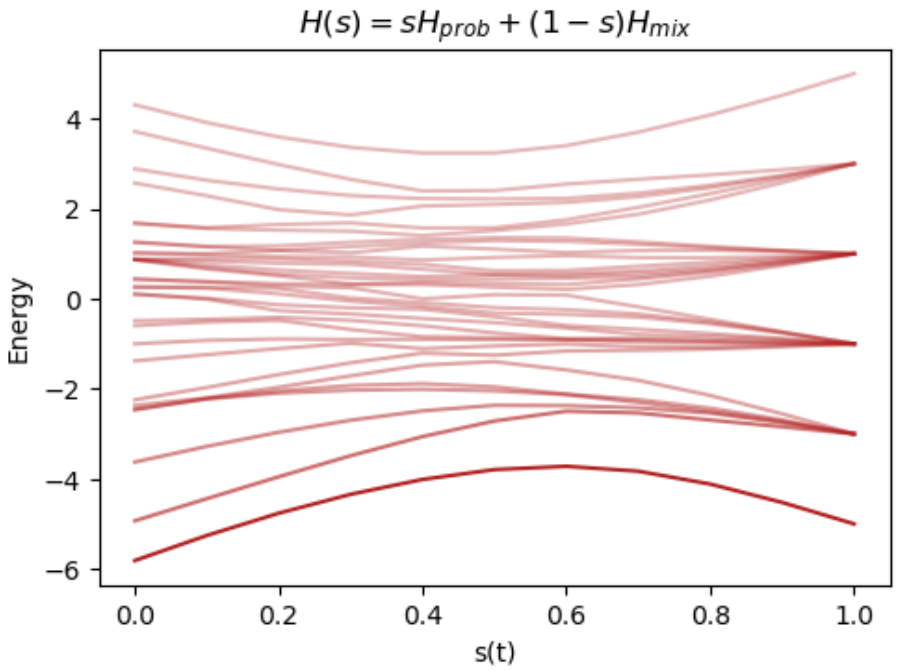
Temperature sweep for **n = 8**

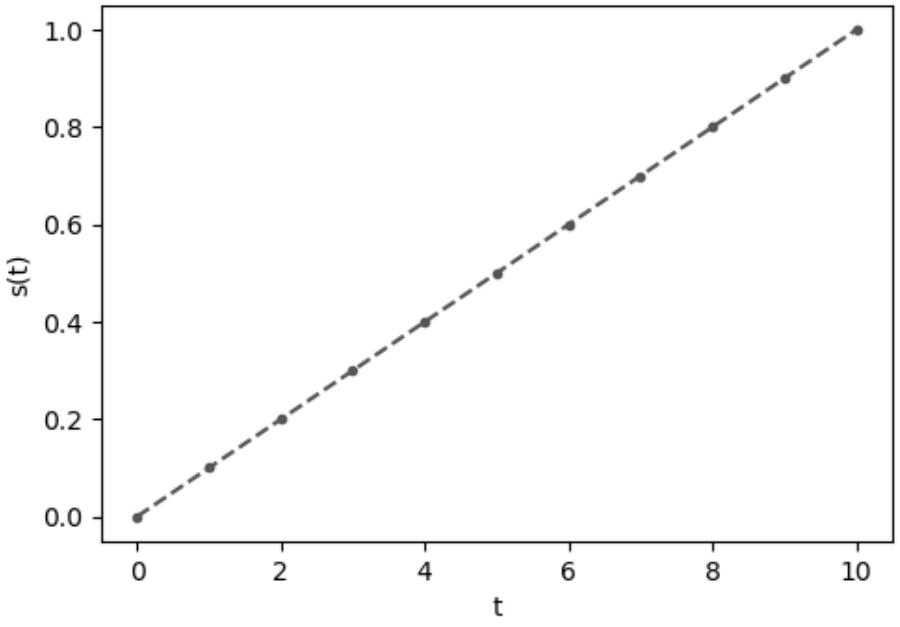


Temperature sweeps for different **n**

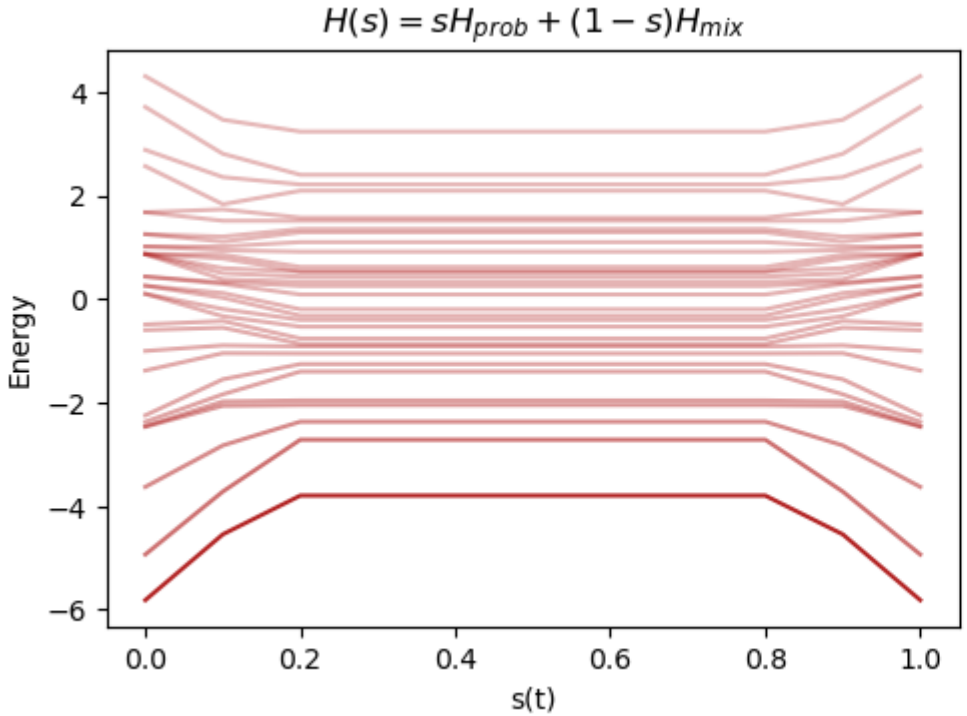


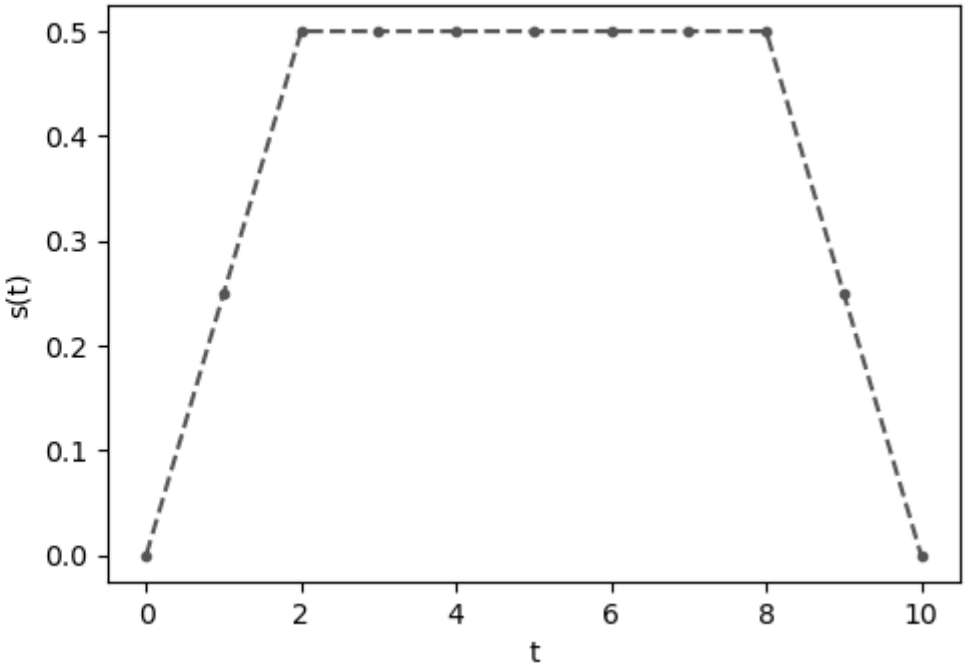
Energy spectrum of the linear schedule





Energy spectrum of reverse annealing





Bottleneck analysis

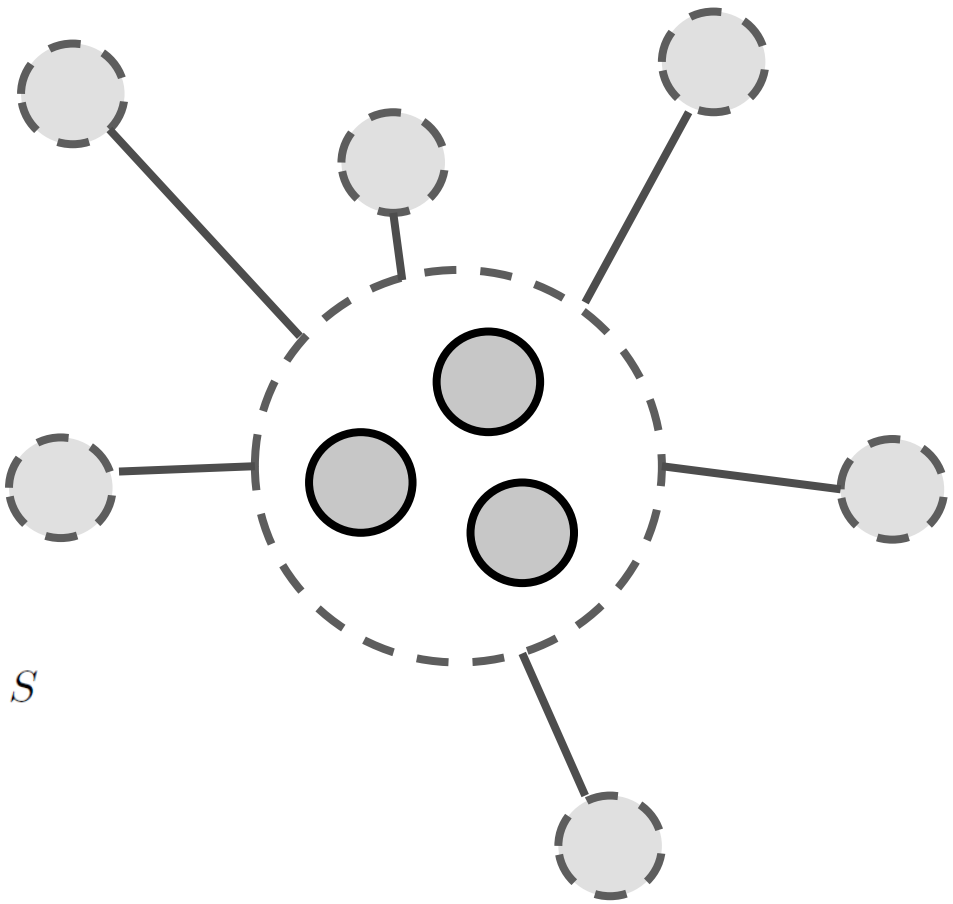
Equilibrium flow: $E(x,y) = \pi(x)P(x,y)$

Flow between subsets: $E(A,B) = \sum_{x\in A, y\in B} E(x,y)$

Bottleneck ratio: $\Phi(A) = \frac{E(A,A^c)}{\pi(A)}$

Bottleneck ratio of the chain: $\Phi_* = \min_A \Phi(A)$

Cheeger's inequality: $\frac{\Phi_*^2}{2} \leq \delta \leq 2\Phi_* \quad \Longrightarrow \quad \delta \leq 2\Phi(A) \ \forall A \in S$



$$
\begin{aligned}
\Phi(A) &= \sum_{x\in A} \frac{\pi(x)}{\pi(A)} \sum_{y\in A^c} P(x,y) \\
&\leq \sum_{x\in A} \frac{\pi(x)}{\pi(A)} \sum_{y\in A^c} Q(x,y) \quad \text{because} \quad A(x,y) \leq 1 \\
&= \sum_{x\in A} \frac{\pi(x)}{\pi(A)} \sum_{y\in A^c} |\langle y|U|x\rangle|^2 \\
&= \ldots \text{ (Time average and Cauchy-Schwartz)} \\
&\leq \sum_{x\in A, y\in A^c} \frac{\pi(x)}{\pi(A)} \sqrt{\mathrm{IPR}(x)\mathrm{IPR}(y)} \quad \text{where} \quad \mathrm{IPR}(x) := \sum_n |\langle n|x\rangle|^4
\end{aligned}
$$

arXiv:2408.07881 **[quant-ph]**